Medical Cost Prediction Using Machine Learning Regression Techniques

Author:
Shaik Shakeer Vali
Aspiring Data Scientist

Keywords: Machine Learning, Healthcare Analytics, Regression, Data Science, Medical Cost Prediction

Abstract

The rapid growth of healthcare expenses has created a significant need for accurate prediction of medical costs. Predicting healthcare expenses can help insurance companies design better policies, reduce financial risk, and support data-driven decision making. Machine learning techniques have become powerful tools for predictive analytics in many fields, including healthcare.

This research focuses on predicting medical insurance costs using machine learning regression techniques. The dataset used in this study includes demographic and lifestyle features such as age, gender, body mass index (BMI), number of children, smoking status, and region. These factors are known to influence healthcare expenses.

Exploratory Data Analysis (EDA) was conducted to understand the distribution of variables and identify relationships between features and medical charges. After preprocessing the dataset, a regression-based machine learning model was developed to estimate healthcare expenses.

The model was evaluated using common regression evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2 score). The model achieved an R^2 score of approximately 0.75, indicating that it can explain a significant portion of the variance in medical charges.

The results highlight the importance of lifestyle factors such as smoking and body mass index in determining healthcare costs. This study demonstrates how machine learning techniques can be applied to healthcare data to improve cost prediction and provide valuable insights for insurance providers and healthcare organizations.

1. Introduction

Healthcare costs have increased significantly in recent decades, making cost prediction an important area of research for insurance companies and healthcare organizations. Accurate prediction of medical expenses can help insurance providers develop fair pricing models and better manage financial risks.

Medical costs are influenced by several factors including age, lifestyle habits, and health conditions. Understanding how these factors affect healthcare expenses is essential for developing predictive models that can estimate insurance costs.

Machine learning has emerged as an effective tool for analyzing large datasets and discovering patterns within data. Unlike traditional statistical methods, machine learning

algorithms can handle complex relationships between variables and generate accurate predictions.

Among various machine learning techniques, regression models are widely used for predicting continuous numerical values such as medical expenses. Regression analysis helps determine the relationship between independent variables and a target variable.

The objective of this study is to develop a regression-based machine learning model that predicts medical insurance charges based on demographic and health-related attributes. The study also aims to identify the most significant factors that influence healthcare costs.

2. Literature Review

Several researchers have applied machine learning techniques to analyze healthcare cost data. Traditional statistical methods such as linear regression have been widely used to model the relationship between patient characteristics and healthcare expenses.

Linear regression is one of the most commonly used algorithms for predicting continuous values. It works by establishing a linear relationship between independent variables and the target variable.

More advanced machine learning algorithms such as Random Forest, Gradient Boosting, and Support Vector Machines have also been used for healthcare cost prediction. These models can capture complex nonlinear relationships within the data.

Despite the availability of advanced algorithms, regression models remain widely used due to their simplicity and interpretability. Regression models allow researchers to understand how each variable contributes to the final prediction.

This research uses a regression-based approach to analyze the relationship between personal attributes and medical insurance charges.

3. Dataset Description

The dataset used in this research is the **Medical Insurance Dataset**, which contains demographic and health-related information about individuals and their corresponding medical insurance costs.

The dataset contains the following variables:

Feature	Description
Age	Age of the individual
Sex	Gender of the individual
BMI	Body Mass Index
Children	Number of dependents

Feature	Description
Smoker	Smoking status
Region	Residential region
Charges	Medical insurance cost

The target variable in this study is **charges**, which represents the medical expenses billed by insurance providers.

Before training the machine learning model, the dataset was examined to ensure data quality. Data preprocessing steps were applied to convert categorical variables into numerical format and prepare the dataset for model training.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of the dataset and identify relationships between variables.

Several visualization techniques were used during the analysis process. Histograms were used to examine the distribution of medical charges, while scatter plots were used to analyze relationships between variables such as age, BMI, and healthcare expenses.

The analysis showed that age has a positive relationship with medical charges. Older individuals generally have higher healthcare expenses compared to younger individuals.

Body Mass Index (BMI) also showed a positive correlation with medical charges. Higher BMI values are often associated with increased health risks, which may result in higher medical expenses.

One of the most significant findings from the exploratory analysis was the effect of smoking status on healthcare costs. Smokers were observed to have significantly higher insurance charges compared to non-smokers.

These insights provided valuable information for building the machine learning model.

-Scatter plots were used to examine relationships between age and medical charges. The visualization shows that medical expenses tend to increase as age increases.

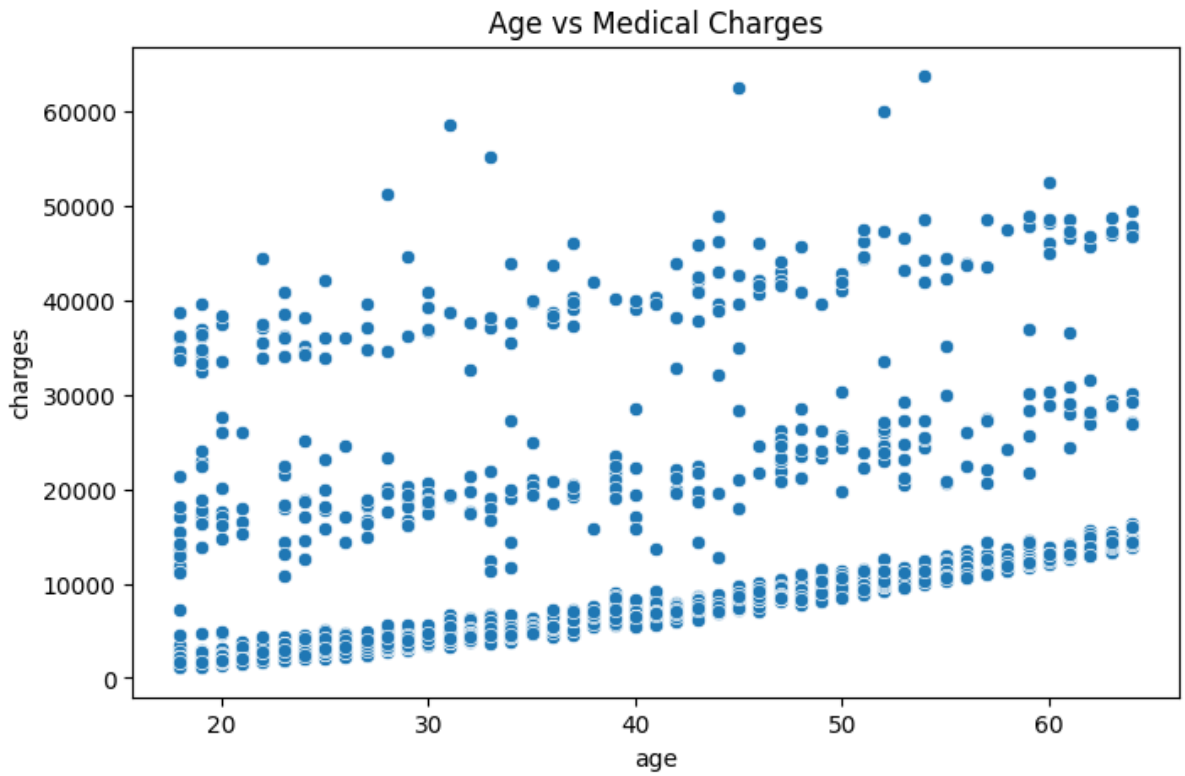

Figure 2: Relationship between BMI and Medical Charges

-Body Mass Index (BMI) was also analyzed to understand its effect on healthcare costs. The scatter plot below illustrates a positive relationship between BMI and medical expenses.

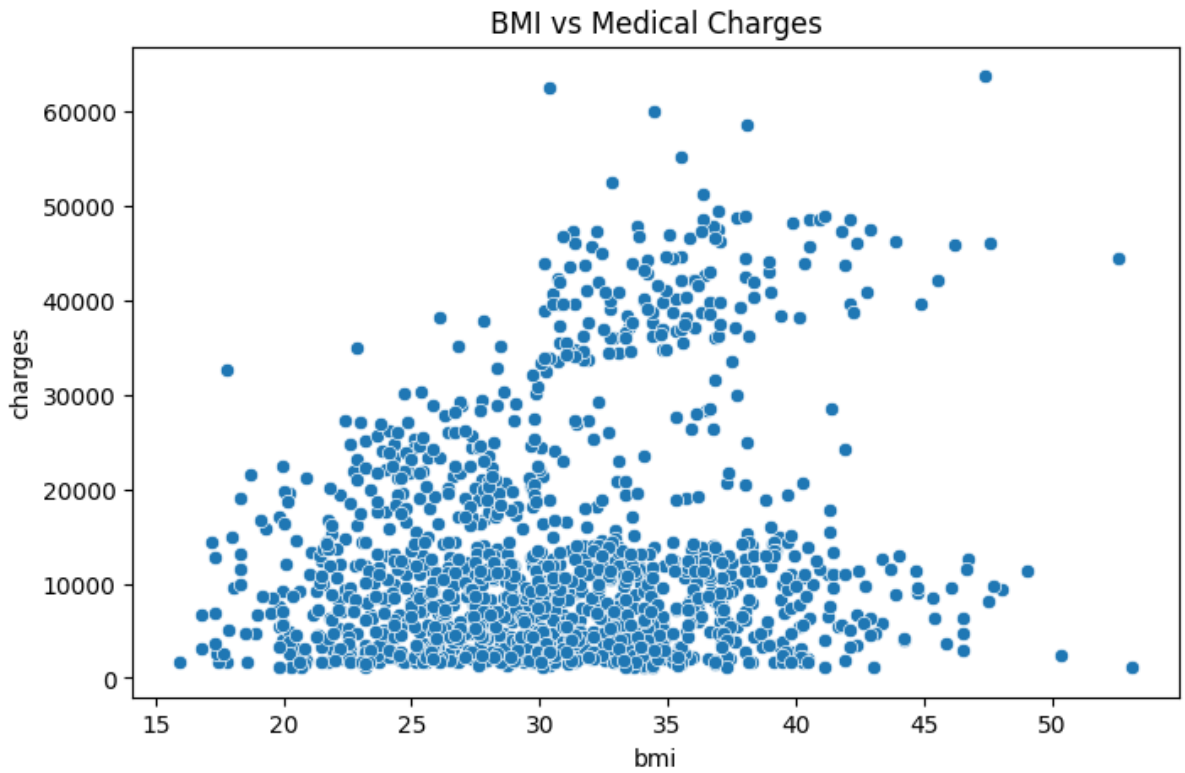

Figure 2: Relationship between BMI and Medical Charges

-A correlation heatmap was generated to analyze relationships between numerical variables in the dataset.

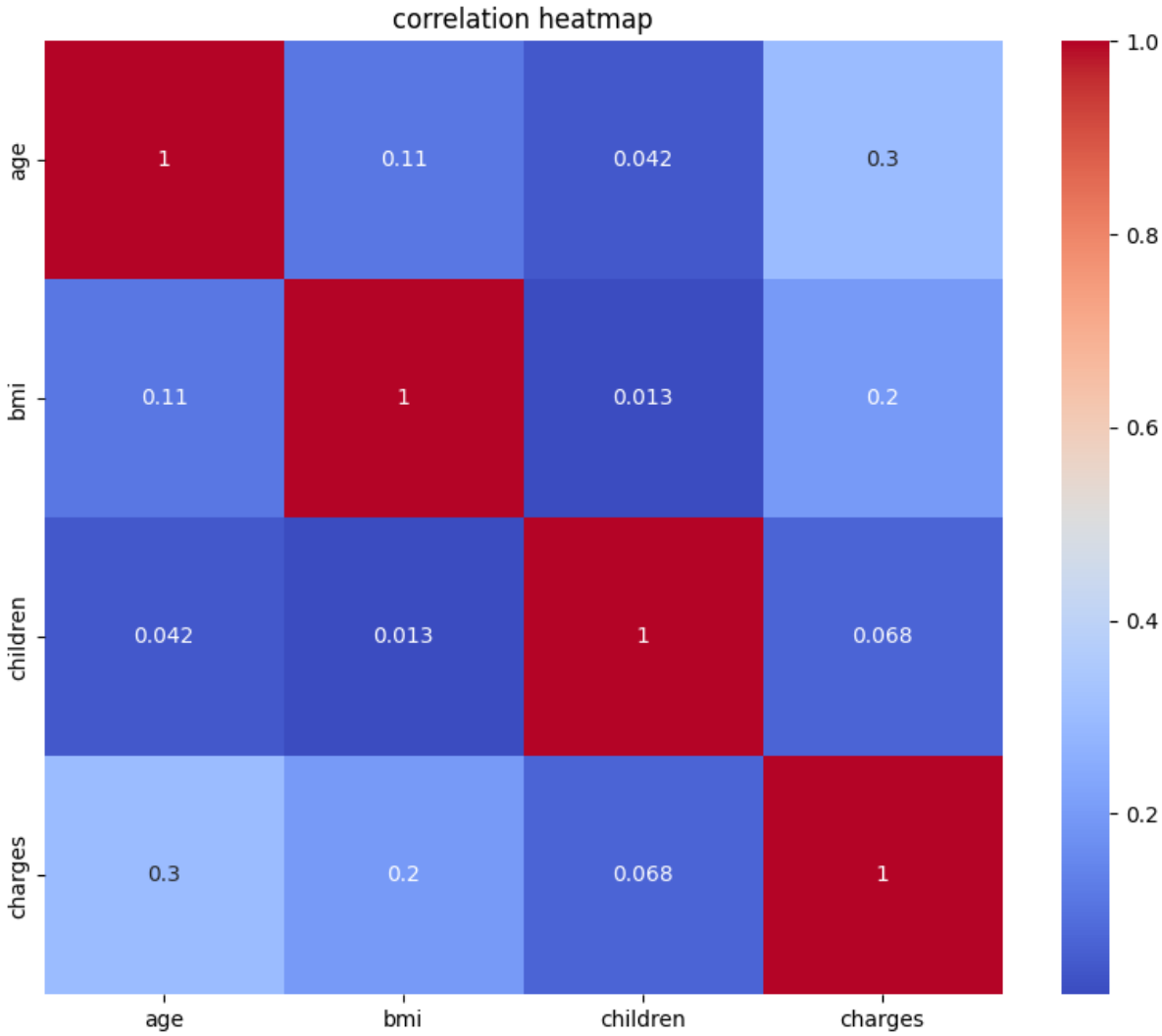

Figure 3: Correlation Heatmap of Dataset Features

-Smoking status has a strong impact on healthcare costs. The boxplot below shows that smokers tend to have significantly higher medical expenses compared to non-smokers.

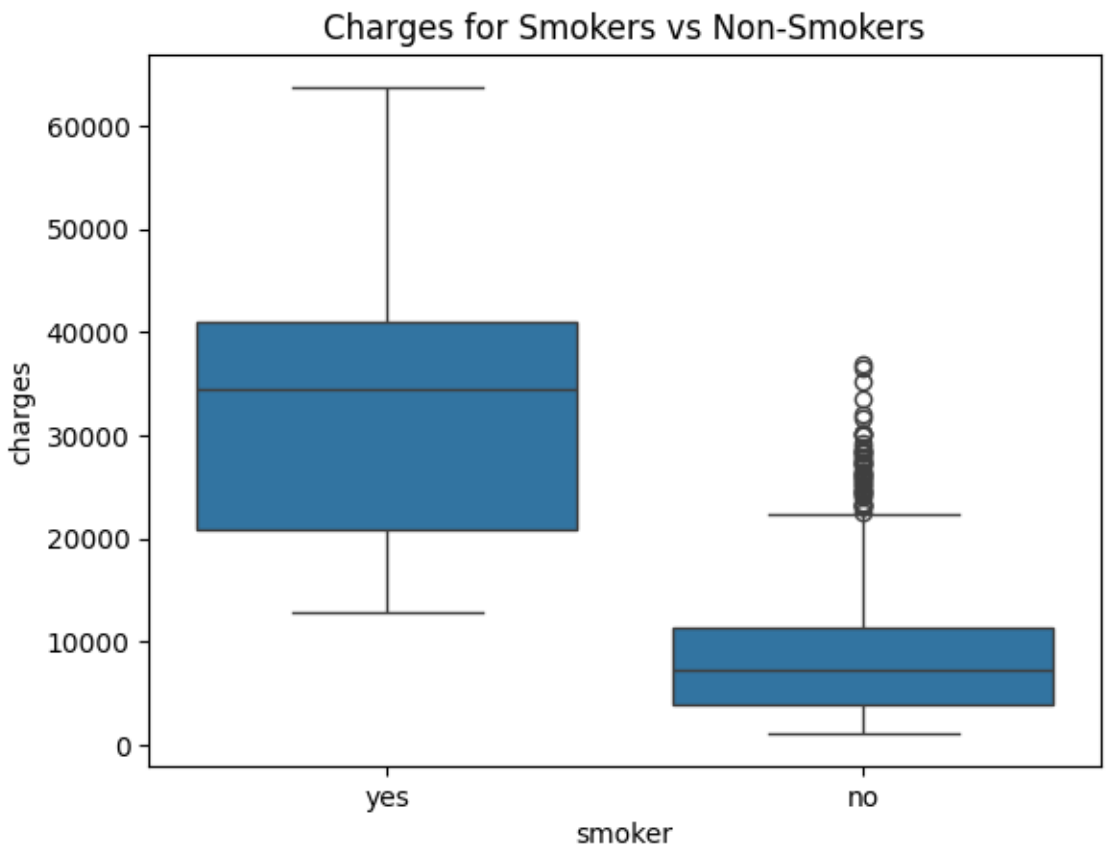

Figure 4: Medical Charges for Smokers vs Non-Smokers

5. Methodology

The methodology used in this research consists of several steps including data preprocessing, feature engineering, model training, and evaluation.

First, the dataset was examined for missing values and inconsistencies. Since the dataset did not contain missing values, no imputation was required.

Categorical variables such as gender, smoker status, and region were converted into numerical values using encoding techniques.

The dataset was then divided into training and testing sets using the train-test split method. This allows the model to be trained on one portion of the dataset and evaluated on unseen data.

The machine learning algorithm used in this study is **Linear Regression**. This algorithm models the relationship between input variables and the target variable using a linear equation.

The regression equation can be expressed as:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$

Where:

- y represents predicted medical charges
- x represents input variables
- β represents model coefficients

6. Model Evaluation

After training the regression model, its performance was evaluated using several metrics commonly used in regression analysis.

The evaluation metrics include:

• Mean Absolute Error (MAE)
• Mean Squared Error (MSE)
• Root Mean Squared Error (RMSE)
• R^2 Score

The R^2 score measures how well the model explains the variance in the target variable. A higher R^2 score indicates better predictive performance.

The regression model achieved an R^2 score of approximately **0.75**, indicating that the model explains about 75% of the variation in medical charges.

7. Results and Discussion

The results of the study indicate that machine learning regression models can effectively predict medical insurance charges.

Smoking status was found to be one of the most important predictors of healthcare costs. Individuals who smoke tend to have significantly higher medical expenses compared to non-smokers.

Body Mass Index also plays an important role in determining healthcare costs. Higher BMI values often correspond to higher medical expenses due to increased health risks.

Age is another factor that contributes to increased healthcare expenses, as older individuals typically require more medical care.

These findings demonstrate the usefulness of machine learning techniques for analyzing healthcare data and identifying key factors influencing medical costs.

8. Conclusion

This study explored the use of machine learning regression techniques for predicting medical insurance costs. The dataset was analyzed using exploratory data analysis, and a regression model was developed to estimate healthcare expenses.

The results demonstrate that regression models can effectively predict medical charges using demographic and lifestyle attributes such as age, BMI, and smoking status.

The study highlights the potential of machine learning techniques in healthcare analytics and demonstrates how predictive models can support decision-making in insurance and healthcare management.

9. Future Work

Future research can improve the prediction accuracy by applying more advanced machine learning algorithms such as Random Forest, Gradient Boosting, and Neural Networks.

Additional features such as medical history, lifestyle habits, and physical activity levels could also be incorporated into the dataset to improve model performance.

Further research may also focus on developing explainable machine learning models that provide deeper insights into healthcare cost prediction.

10. References

1. Medical Insurance Dataset – Kaggle
2. Scikit-learn Machine Learning Library Documentation
3. Research studies on healthcare cost prediction using machine learning